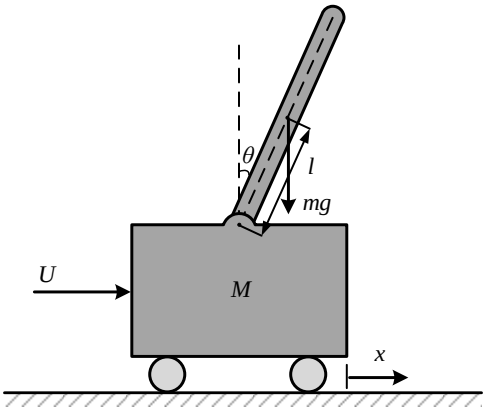
Huazhong University of Science and Technology
School of Mechanical Science and Engineering
System Dynamics
Homework 2

1. Derive the dynamic model of the inverted pendulum;

   Linearize the dynamic model at the equilibrium state ($x = \dot{x} = 0, \theta = \dot{\theta} = 0$) and find the transfer function $\frac{\Theta(s)}{U(s)}$.



Solution:
The dynamic model of the inverted pendulum:

$$(M+m)\ddot{x} + (ml\cos\theta)\ddot{\theta} - ml\dot{\theta}^2\sin\theta = u$$

$$\left(J + ml^2\right)\ddot{\theta} + ml\cos\theta\ddot{x} - mgl\sin\theta = 0$$

At θ=0, the nonlinear terms in the dynamic model can be linearized as follows:

$$(ml\cos\theta)\ddot{\theta} \approx (ml\cos\theta)\ddot{\theta}\big|_{\theta=0} + \frac{d}{d\theta}\left[(ml\cos\theta)\ddot{\theta}\right]\Big|_{\theta=0}\cdot(\theta-0) = ml\ddot{\theta} + \left[(-ml\sin\theta)\ddot{\theta}\right]\Big|_{\theta=0}\cdot(\theta-0) = ml\ddot{\theta}$$

$$ml\dot{\theta}^2\sin\theta \approx \left(ml\dot{\theta}\sin\theta\right)\Big|_{\substack{\theta=0\\ \dot{\theta}=0}} + \frac{\partial}{\partial\theta}\left[ml\dot{\theta}^2\sin\theta\right]\Big|_{\substack{\theta=0\\ \dot{\theta}=0}}\cdot(\theta-0) + \frac{\partial}{\partial\dot{\theta}}\left[ml\dot{\theta}^2\sin\theta\right]\Big|_{\substack{\theta=0\\ \dot{\theta}=0}}\cdot\left(\dot{\theta}-0\right)$$

$$= \left[\left(ml\dot{\theta}^2\cos\theta\right)\right]\Big|_{\substack{\theta=0\\ \dot{\theta}=0}}\cdot(\theta-0) + \left[\left(2ml\dot{\theta}\sin\theta\right)\right]\Big|_{\substack{\theta=0\\ \dot{\theta}=0}}\cdot\left(\dot{\theta}-0\right) = 0$$

$$ml\cos\theta\ddot{x} \approx (ml\cos\theta)\ddot{x}\big|_{\theta=0} + \frac{d}{d\theta}\left[(ml\cos\theta)\ddot{x}\right]\Big|_{\theta=0}\cdot(\theta-0) = ml\ddot{x} + \left[(-ml\sin\theta)\ddot{x}\right]\Big|_{\theta=0}\cdot(\theta-0) = ml\ddot{x}$$

$$mgl\sin\theta \approx (mgl\sin\theta)\big|_{\theta=0} + \frac{d}{d\theta}\left[(mgl\sin\theta)\right]\Big|_{\theta=0}\cdot(\theta-0) = \left[(mgl\cos\theta)\right]\Big|_{\theta=0}\cdot(\theta-0) = mgl\theta$$

Therefore, the linearized model can be derived as

$$(M+m)\ddot{x} + ml\ddot{\theta} = u$$

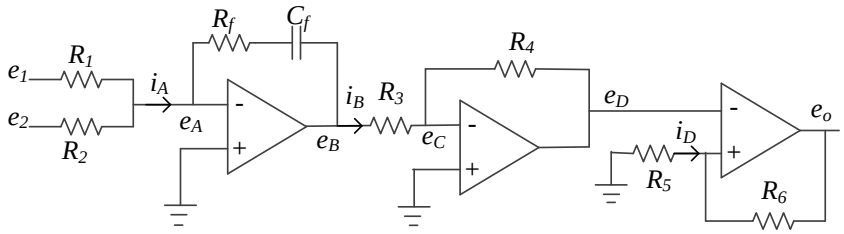
$$\left(J + ml^2\right)\ddot{\theta} + ml\ddot{x} - mgl\theta = 0$$

Taking Laplace transform and eliminating X(s), yields

$$\frac{\Theta(s)}{F(s)} = \frac{-ml}{(MJ + mJ + Mml^2)s^2 - (M+m)mgl}$$

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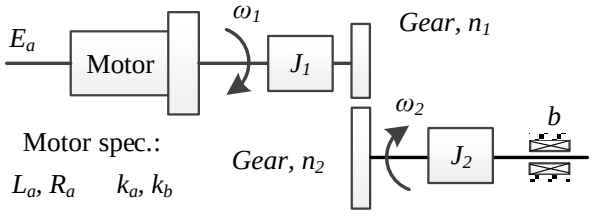
2. Derive the transfer function between E_o and the inputs E_1, $E2$:



Solution：

$$E_o = -\left(\frac{R_4(R_5+R_6)R_f}{R_3R_5} + \frac{R_4(R_5+R_6)}{C_fR_3R_5s}\right)\left(\frac{E_1}{R_1}+\frac{E_2}{R_2}\right)$$

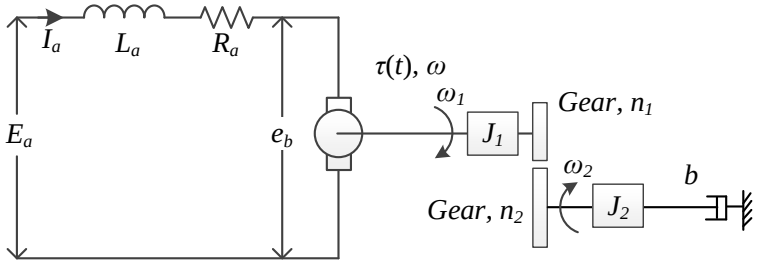
3. For the system shown to the right, derive the transfer function between the input voltage and the rotational speed of J_2:



(a) Draw the free-body diagram

(b) Find $\dfrac{\Omega_2(s)}{E_a(s)}$

Solution:



$$\frac{\Omega_2(s)}{E_a(s)} = \frac{k_aA}{\left(J_1+J_2A^2\right)L_as^2+\left(J_1R_a+J_2A^2R_a+L_abA^2\right)s+R_abA^2+k_ak_b}$$, where $A = n_1/n_2$